LOMBA KOMPETENSI SISWA
SMK TINGKAT NASIONAL KE-29
TAHUN TAHUN 2021



PRE-TEST PROJECT
MODUL A – DAY 2 CLIENT SERVER

BIDANG LOMBA

TEKNOLOGI INFORMASI SISTEM ADMINISTRASI JARINGAN

IT NETWORK SYSTEMS ADMINISTRATION

DESCRIPTION OF PROJECT

A small startup company's production environment contains numerous services within multiple Operating Systems. We will ask you to install, configure, and integrate the different services. You will have two work days to complete two different sets of tasks. Each set of tasks is to be completed within one work day. For the second work day, you will continue the work of another engineer, who should have fully configured systems according to the day one set of tasks. You may check, change, or update the existing systems, but we will only score the tasks you're supposed to do on the second work day.

DAY 2 - LINUX SERVICES

Folder Backup

- Backup all users virtual host folder in srv1 and srv2 to C:\public in HYPERV1. Backup all 100 users folder.
    - /var/www/user001/
    - /var/www/user002/
    - /var/www/user003/
    - …
    - /var/www/user099/
    - /var/www/user100/
- Mount the remote backup folder locally at /backup.
    - For server srv1, place backup content in directory /backup/srv1/
    - For server srv2, place backup content in directory /backup/srv2/
- Schedule backup every 1 hour using cron as root user. It is okay to overwrite previous backups.

Email Services

- Install any mail service and configure a sending-only mail server in srvind.indonesia.com
    - Secure SMTP Port 465
    - Enable PAM Authentication
        - Make sure all local users are able to authenticate
    - Use the self-signed certificate from Windows CA for this task.

Securing HTTP Access

- Use CA from Windows to generate self-signed certificates required for this task.
- Enable HTTPS in the load balancer serving www.sabang.net.
- Enable HTTPS in the web server serving user websites.
    - Configure load balancer to forward traffic non-terminated HTTPS. Load balancer will not handle HTTPS, HTTPS is handled by the backend web server.
    - Only enable to the first 5 and last 5 users :
        - user001.public.sabang.net
        - user002.public.sabang.net
        - user003.public.sabang.net
        - user004.public.sabang.net
        - user005.public.sabang.net
        - user096.public.sabang.net
        - user097.public.sabang.net
        - user098.public.sabang.net
        - user099.public.sabang.net
        - user100.public.sabang.net

DAY 2 - WINDOWS SERVICES

File Sharing

- Create and Share Folder C:\backup\ in host HYPERV1
  - Enable AD Authentication, permit user 'Admnistrator' only
- Create and Share Folder C:\public\ in host HYPERV1
  - Disable Authentication, anonymous user can read and write to this folder.

Backup Configuration

- Access DC and Schedule Backup using Windows Backup.
- Backup folder C:\cert and all its contents to \\HYPERV1\backup and \\HYPERV2\backup right away.
- Schedule the backup every day at 1 AM.
- Both backup targets should have identical content.

Cross Platform iSCSI

- Configure iSCSI Initiator on HYPERV1 and HYPERV2
- Connect to 10 disks in Sabang iSCSI.
  - HYPERV1 will connect to srv1
  - HYPERV2 will connect to srv2
- Create an NTFS filesystem and mount all 10 disks to E:/, F:/, and so on in any order.
  - Make sure 10 extra disks is accessible via File Explorer

Virtualization Platform

- Enable Hyper-V features in HYPERV1 and HYPERV2
- Configure Hyper-V to be able to run Virtual Machines successfully.
- Configure and enable 2-node Failover Clustering in the hosts above.
- Ensure Live Migration feature is supported
- Create VM with name 'Cirros' from disk 'cirros.iso' attached in HYPERV1.

Configure Web Service HTTPS

- Access HYPERV1 and HYPERV2 and Install IIS web service.
- Both servers will host the same content. Failover will be enabled so when one server is down, service is still available.
- Serve web https://www.merauke.net
  - Serve file index.html as specified in the appendix
  - Use a Self-Signed Certificate from LKSN2021-CA to enable HTTPS.
- Serve 25 user websites:
  - Make sure it is accessible using following URL https://user01.merauke.net, https://user02.merauke.net, https://user03.merauke.net, and so on until https://user25.merauke.net.
  - Use a Self-Signed Certificate from LKSN2021-CA to enable HTTPS.
  - Serve file users.html in the following directory:
    - C:\www\users\01\users.html
    - C:\www\users\02\users.html
    - …
    - C:\www\users\25\users.html

Configure Web Failover

- Enable Failover in HYPERV1 and HYPERV2 using NLB
    - User Virtual IP Address 172.17.11.100
    - Make sure all websites are accessible when one of the servers is DOWN.

Setup VPN

- Access HYPERV2 and Install RRAS and NPS.
- Configure VPN Server for Windows Client
    - Authenticate using username and password in Active Directory.
    - Permit any users in the domain to authenticate.
- Make sure budi-pc can connect to the VPN using previously configured Public IP from NAT.
    - Access the network device and open the ports needed.

DAY 2 - NETWORK INFRASTRUCTURE

Security Configuration

- Enable firewall in routers to protect sabang.net, indonesia.com, and merauke.net network.
- Restrict all access to the protected network by default.
- Permit access from merauke.net to sabang.net via VPN.
- Permit access from sabang.net to merauke.net via VPN.
- Permit access from merauke.net to the public addresses.
- Permit access from sabang.net to the public addresses.
- Permit access from indonesia.com to the public addresses.
- Permit access from all networks to TCP port 465 to srvind.indonesia.com

Secure Virtual Private Network

- Configure VPN on the routers to enable tunnelling between sabang.net and merauke.net.
- Secure the VPN with ikev2 protocol.
- Allow devices in sabang.net and merauke.net to communicate via encrypted VPN tunnel.

APPENDIX

Users and Passwords

Username	Password
patah	KesultananDemak2021

Network Devices Addressing Table

Device Name	IP Address CIDR	Network
RTA	172.17.1.2/30	sabang.net
	192.168.199.1/21	indonesia.com
	30.9.65.1/30	Public
ISP	30.9.65.2/30	Public
	30.9.19.1/24	Public
	30.9.65.9/29	Public
RTB-1	30.9.65.11/29	Public
	172.17.9.1/22	merauke.net
RTB-2	30.9.65.12/29	Public
	172.17.9.2/22	merauke.net

End Devices Addressing Table

Device Name	IP Address CIDR	Network
srv1.sabang.net	172.16.1.201/20	sabang.net
srv2.sabang.net	172.16.1.202/20	sabang.net
ani-pc	172.16.1.101/20	sabang.net
fw.sabang.net	172.16.15.254/20	sabang.net

	172.17.1.1/30	sabang.net
srvind.indonesia.com	192.168.192.168/21	srvind.indonesia.com
DC	172.17.10.100/22	merauke.net
HYPERV1	172.17.11.101/22	merauke.net
HYPERV2	172.17.11.102/22	merauke.net
budi-pc	30.9.19.10/24	Public

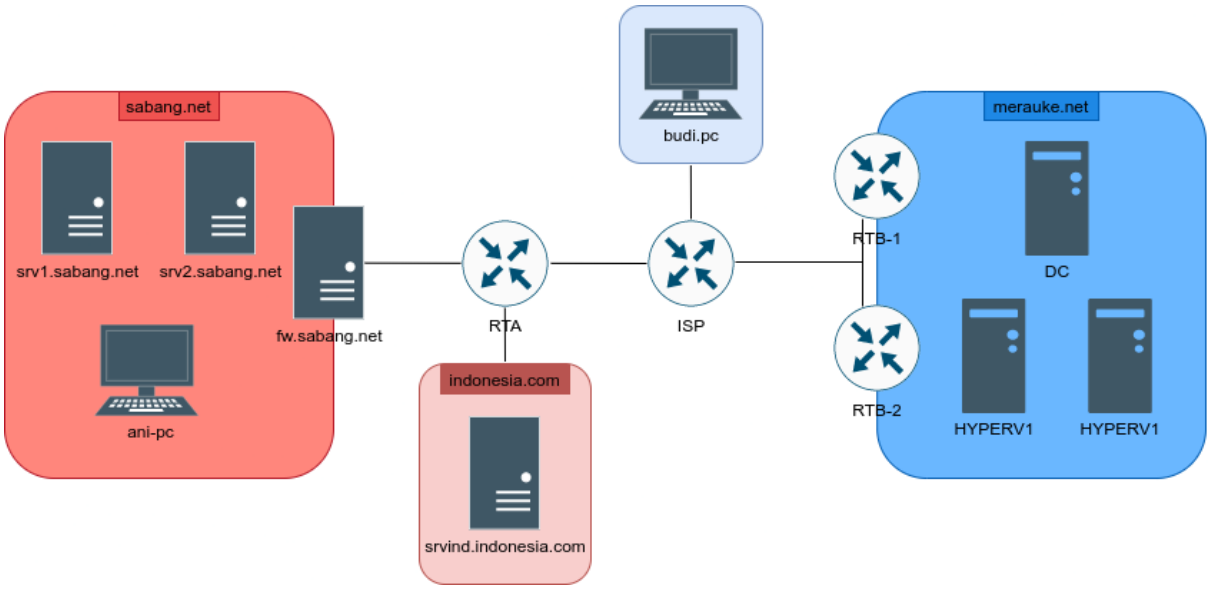
index.html

```
<h1> Default Page </h1><br>
<br>
This page has not been modified by the owners.
```

users.html

```
<h1> Users Page </h1><br>
<br>
This is the default page for users' websites.
```

TOPOLOGY


sabang.net
srv1.sabang.net
srv2.sabang.net
fw.sabang.net
ani-pc
RTA
budi.pc
ISP
indonesia.com
srvind.indonesia.com
merauke.net
RTB-1
RTB-2
DC
HYPERV1
HYPERV1